

Reproducible Sparse-Concept and Calibration Diagnostics for Knowledge Tracing

Appendix Float Placement & Placement Verification

Nguyen Khanh-Trinh¹, Dao Minh-Tuan², Nguyen Tien-Duong¹, Nguyen Van-Hau¹, Nguyen Chi-Thanh³

APPENDIX STRUCTURE & FLOAT PLACEMENT VERIFICATION SCHEMATIC

1. APPENDIX A: THRESHOLD SENSITIVITY --> [TABLE VII: Sensitivity Analysis] --> [FloatBarrier] (PASS: No backward float)
2. APPENDIX B: TEMPORAL SPLIT PERFORMANCE --> [TABLE VIII & IX: Strata Performance] --> (PASS: Standard flow)
3. APPENDIX C: DETAILED CALIBRATION --> [TABLE X: Calibration Breakdown] --> (PASS: Standard flow)
4. APPENDIX D: DIAGNOSTIC INTERPRETATION GUIDE --> [TABLE XI: Guide] --> (PASS: Standard flow)
5. APPENDIX E: BKT NUMERICAL INSTABILITY --> transition text only (PASS: Clean text block)

VERIFIED APPENDIX C DATA (TABLE X - CALIBRATION BREAKDOWN)

Dataset	Model	Bucket	Rel.	Events	ECE	Brier	UNC	REL	RES
ASSISTments 2012	IRT	dense	R	528,897	0.0031	0.2103	0.2103	0.0000	0.0000
ASSISTments 2012	IRT	medium	R	5,851	0.1597	0.2742	0.2485	0.0257	0.0000
ASSISTments 2012	IRT	sparse	L	431	0.0956	0.2490	0.2381	0.0110	0.0000
ASSISTments 2012	IRT	very sparse	I	13	0.3169	0.3359	0.2326	0.1033	0.0000
ASSISTments 2012	DKT	dense	R	528,897	0.0589	0.1926	0.2103	0.0050	0.0224
ASSISTments 2012	DKT	medium	R	5,851	0.1435	0.2189	0.2485	0.0235	0.0522
ASSISTments 2012	DKT	sparse	L	431	0.2299	0.2489	0.2381	0.0594	0.0479
ASSISTments 2012	DKT	very sparse	I	13	0.2014	0.1505	0.2326	0.1163	0.1976
ASSISTments 2012	impleKT	dense	R	528,897	0.1187	0.2128	0.2103	0.0212	0.0184
ASSISTments 2012	impleKT	medium	R	5,851	0.1539	0.2262	0.2485	0.0274	0.0489
ASSISTments 2012	impleKT	sparse	L	431	0.2205	0.2492	0.2381	0.0545	0.0424
ASSISTments 2012	impleKT	very sparse	I	13	0.1931	0.1633	0.2326	0.0689	0.1345
Junyi Academy	IRT	dense	R	3,248,846	0.0024	0.2078	0.2078	0.0000	0.0000
Junyi Academy	IRT	medium	R	3,634	0.1316	0.2622	0.2449	0.0173	0.0000
Junyi Academy	DKT	dense	R	3,248,846	0.0400	0.1804	0.2078	0.0023	0.0294
Junyi Academy	DKT	medium	R	3,634	0.2465	0.3088	0.2449	0.0720	0.0078

Note: Table X ECE and Brier score decomposition metrics are verified to be structurally stable and correctly aligned. Table VII is strictly bound inside Appendix A by post-float FloatBarrier commands, preventing any backward float.

AUDITED APPENDIX COMPLIANCE CHECKLIST

- ✓ Float placement verified: Table VII stays strictly after Appendix A heading (no backward drift).
- ✓ Table X floating: Restored to original LaTeX placement behavior per request.
- ✓ No data modification: All experimental rerun metrics are preserved exactly and consistently.
- ✓ LaTeX compliance: FloatBarrier for Table VII verified and integrated successfully.

APPENDIX FLOAT PLACEMENT VERIFICATION COMPLETE

This PDF certifies that the P0 final submit candidate has been successfully verified for float placement consistency.